

Unit-3

Second Order Linear Differential Equations

Homogenous Equation with Constant Coefficients

Definition

An ordinary second order differential equation is of the form

$$\frac{d^2y}{dt^2} = F\left(t, y, \frac{dy}{dt}\right) \quad \dots \quad (1)$$

Linear and Non-linear Second Order ODE

Equation (1) is said to be linear if the function $F\left(t, y, \frac{dy}{dt}\right)$ has the form

$$F\left(t, y, \frac{dy}{dt}\right) = g(t) - p(t)\frac{dy}{dt} - q(t)y \quad \dots \quad (2),$$

that is, F is linear in y and $\frac{dy}{dt}$.

In equation (2), g, p, q are constants or functions of independent variable t alone (but not of y).

In this case the equation (2) can be written as

$$\frac{d^2y}{dt^2} = g(t) - p(t)\frac{dy}{dt} - q(t)y$$

or, $y'' + p(t)y' + q(t)y = g(t) \quad \dots \quad (3)$

The equation (1) is said to be non-linear if it is not linear, that is, if it is not of the form (3).

An initial value problem (IVP) of second order is the differential equation

$$y'' + p(t)y' + q(t)y = g(t)$$

with initial conditions

$$y(t_0) = y_0 \text{ and } y'(t_0) = y'_0 \quad \dots \quad (4)$$

Homogenous and Non-homogenous Second Order ODE

A second order linear ode

$$y'' + p(t)y' + q(t)y = g(t)$$

is said to be homogenous if $g(t) = 0$ for all t .

In this case the equation takes the form

$$y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (5)$$

The equation (3) is said to be non-homogenous if it is not homogenous, i.e. $g(t) \neq 0$ for some t .

Homogenous Equation with Constant Coefficients

The homogenous linear differential equation

$y'' + p(t)y' + q(t)y = 0$ is said to be homogeneous linear ODE with constant coefficients if $p(t)$ and $q(t)$ are constants. If $p(t)$ and $q(t)$ are constants, it takes the form

$$ay'' + by' + cy = 0 \quad \dots \quad (6)$$

where a, b, c are constants.

Solution of (6)

Let $y = e^{mt}$ be a solution of (6), then

$$y' = me^{mt} \text{ and } y'' = m^2e^{mt}.$$

Substituting these values in (6) we get

$$am^2e^{mt} + bme^{mt} + ce^{mt} = 0$$

$$\text{or, } (am^2 + bm + c)e^{mt} = 0$$

Since $e^{mt} \neq 0 \forall t \in R$, we have

$$am^2 + bm + c = 0 \quad \dots \quad (7)$$

Here, the equation (7) is called the characteristic equation or auxiliary equation (A.E.) of the differential equation (6).

There are three different possible cases:

Case I [When the roots of AE are real and distinct]

If $m = \alpha, \beta$ are two real and distinct roots of (7), then $e^{\alpha t}$ and $e^{\beta t}$ are two solutions of (6).

Then, the general solution of (6) is

$$y = c_1 e^{\alpha t} + c_2 e^{\beta t},$$

where c_1 and c_2 are arbitrary constants.

Case II [When the roots of A.E. are real and equal]

Let $m = \alpha = \beta$ be two equal and real roots of A.E. Then, the general solution of (6) is

$$y = (c_1 + c_2 t) e^{\alpha t}.$$

Case III [When the roots of AE are imaginary]

Let $m = \alpha + i\beta$ and $m = \alpha - i\beta$ be two imaginary (complex conjugates) roots of A.E. Then, the general solution of (6) is

$$y = (c_1 \cos \beta t + c_2 i \sin \beta t) e^{\alpha t}$$

Example 1. Find the general solution of the following differential equation $y'' - 9y' + 9y = 0$

Solution

The given differential equation is

$$y'' - 9y' + 9y = 0 \quad \dots \quad (1)$$

The A.E. of (1) is

$$\begin{aligned} m^2 - 9m + 9 &= 0 \\ \therefore m &= \frac{-(-9) \pm \sqrt{(-9)^2 - 4 \cdot 1 \cdot 9}}{2 \cdot 1} \\ &= \frac{9 \pm \sqrt{81 - 36}}{2} = \frac{9 \pm \sqrt{45}}{2} = \frac{9 \pm 3\sqrt{5}}{2} \\ \therefore m &= \frac{9}{2} + \frac{3\sqrt{5}}{2}, \frac{9}{2} - \frac{3\sqrt{5}}{2} \end{aligned}$$

Hence, the general solution is

$$y = c_1 e^{\left(\frac{9+3\sqrt{5}}{2}\right)t} + c_2 e^{\left(\frac{9-3\sqrt{5}}{2}\right)t}.$$

Example 2. Find the solution of the following IVP:

$$6y'' - 5y' + y = 0, y(0) = 4 \text{ and } y'(0) = 0.$$

Solution

The given differential equation is

$$6y'' - 5y' + y = 0 \quad \dots \quad (1)$$

and the initial conditions are

$$y(0) = 4 \text{ and } y'(0) = 0 \quad \dots \quad (2)$$

The A.E. of (1) is

$$6m^2 - 5m + 1 = 0 \quad \dots \quad (3)$$

$$\text{or, } 6m^2 - 3m - 2m + 1 = 0$$

$$\text{or, } (3m - 1)(2m - 1) = 0$$

$$\therefore m = \frac{1}{2}, \frac{1}{3}$$

The general solution of (1) is

$$y = c_1 e^{\frac{1}{2}t} + c_2 e^{\frac{1}{3}t} \quad \dots \quad (4)$$

Differentiating it w.r.t. t , we get

$$\frac{dy}{dt} = \frac{1}{2}c_1 e^{\frac{1}{2}t} + \frac{1}{3}c_2 e^{\frac{1}{3}t}$$

$$\text{or, } y' = \frac{c_1}{2} e^{\frac{1}{2}t} + \frac{c_2}{3} e^{\frac{1}{3}t} \quad \dots \quad (5)$$

Using $y(0) = 4$ in (4) we get

$$c_1 + c_2 = 4 \quad \dots \quad (i)$$

and using $y'(0) = 0$ in (5), we get

$$\frac{c_1}{2} + \frac{c_2}{3} = 0 \Rightarrow 3c_1 + 2c_2 = 0 \quad \dots \quad (ii)$$

Using (i) in (ii), we get

$$3(4 - c_2) + 2c_2 = 0$$

$$\text{or, } 12 - 3c_2 + 2c_2 = 0$$

$$\text{or, } 12 - c_2 = 0, \quad \therefore c_2 = 12$$

And

$$c_1 = 4 - 12 = -8$$

Hence, the required solution of IVP is

$$y = -8e^{\frac{t}{2}} + 12e^{\frac{t}{3}}.$$

Solution of Linear Homogeneous Equations: Wronskian

The linear homogeneous differential equation of second order is

$$y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1),$$

where p and q are continuous real valued functions in an open interval $I = (\alpha, \beta)$ with finite or infinite length.

Let ϕ be any function which is twice differentiable in I . We now define a differential operator L by

$$L[\phi] = \phi'' + p\phi' + q\phi \quad \dots \quad (2)$$

The value of $L[\phi]$ at t is

$$L[\phi](t) = \phi''(t) + p\phi'(t) + q\phi(t)$$

OR

$$L[\phi](t) = \phi''(t) + p(t)\phi'(t) + q(t)\phi(t)$$

From (2)

$$L[\phi] = D^2\phi + pD\phi + q\phi$$

$$\text{or, } L[\phi] = (D^2 + pD + q)(\phi)$$

Where $D = \frac{d}{dt}$, is called the derivative operator.

$$\therefore L = D^2 + pD + q.$$

Taking $\phi(t) = y$, we have

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (3)$$

Along with equation (3), we consider the initial conditions

$$y(t_0) = y_0 \text{ and } y'(t_0) = y'_0 \quad \dots \quad (4),$$

where $t_0 \in I = (\alpha, \beta)$ and y_0 and y'_0 are given real numbers.

Theorem 1 (Existence and Uniqueness Theorem)

The initial value problem (IVP)

$$y'' + p(t)y' + q(t)y = g(t), y(t_0) = y_0 \text{ and } y'(t_0) = y'_0$$

where p, q, g are continuous on an open interval I that contains the point t_0 has exactly one solution $y = \phi(t)$ and the solution exists throughout the internal I .

Example 1. Find the largest interval in which the solution of the IVP

$$(t^2 - 3t)y'' + ty' - (t + 3)y = 0, y(1) = 2, y'(1) = 1$$

is certain to exist.

Solution

The given IVP is

$$(t^2 - 3t)y'' + ty' - (t + 3)y = 0 \quad \dots \quad (1)$$

with initial conditions

$$, y(1) = 2, y'(1) = 1 \quad \dots \quad (2)$$

Now, (1) can be written as

$$y'' + \frac{t}{(t^2 - 3t)}y' - \frac{(t + 3)}{(t^2 - 3t)}y = 0$$

or,

$$y'' + \frac{1}{(t - 3)}y' - \frac{(t + 3)}{t(t - 3)}y = 0 \quad \dots \quad (3)$$

Comparing it with

$$y'' + p(t)y' + q(t)y = g(t),$$

we have

$$p(t) = \frac{1}{(t - 3)}, q(t) = \frac{(t + 3)}{t(t - 3)}, g(t) = 0$$

Thus, the function $p(t)$ and $q(t)$ are discontinuous at point $t = 0$ and $t = 3$. Moreover, the initial point $t_0 = 1$.

So, the longest interval which contains $t_0 = 1$ and $p(t), q(t)$ are continuous is $(0, 3)$.

Therefore, by the theorem, $(0, 3)$ is the longest interval in which the solution exists.

Theorem 2 (Superposition Principle)

Let y_1 and y_2 be two solutions of the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0.$$

Then, the linear combination $c_1y_1 + c_2y_2$ is also a solution for any values of c_1 and c_2 .

Proof

The given differential equation is

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

and let

$$y = c_1y_1(t) + c_2y_2(t) \quad \dots \quad (2),$$

where y_1 and y_2 are solutions of (1) and c_1, c_2 are arbitrary constants.

Then,

$$\begin{aligned} L[y] &= L[c_1y_1 + c_2y_2] \\ &= [c_1y_1 + c_2y_2]'' + p(t)[c_1y_1 + c_2y_2]' + q(t)[c_1y_1 + c_2y_2] \\ &= c_1[y_1'' + p(t)y_1' + q(t)y_1] + c_2[y_2'' + p(t)y_2' + q(t)y_2] \\ &= c_1L[y_1] + c_2L[y_2] \end{aligned}$$

This shows that the differential operator L is linear.

Since y_1 and y_2 are solutions of $L[y] = 0$, we have

$$L[y_1] = 0 \text{ and } L[y_2] = 0$$

Hence,

$$\begin{aligned} L[c_1y_1 + c_2y_2] &= c_1L[y_1] + c_2L[y_2] \\ &= c_1 \cdot 0 + c_2 \cdot 0 = 0 \end{aligned}$$

Therefore, $y(t) = c_1y_1 + c_2y_2$ is also a solution of $L[y] = 0$.

This proves the theorem.

Consider the initial value problem

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

with $y(t_0) = y_0$ and $y'(t_0) = y_0$,

Using $y(t_0) = y_0$

$$y = c_1 y_1 + c_2 y_2 \quad \dots \quad (2)$$

reduces to

$$y_0 = c_1 y_1(t_0) + c_2 y_2(t_0) \quad \dots \quad (3)$$

Differentiating (2), we get

$$y' = c_1 y_1' + c_2 y_2'$$

Using $y'(t_0) = y_0$ in it, we get

$$y_0' = c_1 y_1'(t_0) + c_2 y_2'(t_0) \quad \dots \quad (4)$$

The determinant of the coefficients of equation (3) and (4) is

$$W = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix} = y_1(t_0)y_2'(t_0) - y_2(t_0)y_1'(t_0) = y_1 y_2' - y_2 y_1'$$

When $W \neq 0$, the equation (3) and (4) have a unique solution c_1 and c_2 . By Cramer's rule, these solutions are given by

$$c_1 = \frac{\begin{vmatrix} y_0 & y_2(t_0) \\ y_0' & y_2'(t_0) \end{vmatrix}}{W}, c_2 = \frac{\begin{vmatrix} y_1(t_0) & y_0 \\ y_1'(t_0) & y_0' \end{vmatrix}}{W}$$

With these values of c_1 and c_2 the expression (2) satisfies the differential equation (1) and the initial conditions $y(t_0) = y_0$ and $y'(t_0) = y_0$.

When $W = 0$, then the equation (3) and (4) have no solutions.

The determinant

$$W = W(y_1, y_2) = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix}$$

is called the Wronskian determinant (or simply wronskian) of the two solutions y_1 and y_2 .

Theorem 3 Let y_1 and y_2 be two solutions of the equation

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

and the initial conditions

$$y(t_0) = y_0 \text{ and } y'(t_0) = y_0' \quad \dots \quad (2)$$

be assigned. Then, there exist constants c_1 and c_2 such that

$$y = c_1 y_1(t) + c_2 y_2(t)$$

satisfies the differential equation (1) and the initial conditions (2)

iff the Wronskian $W = \begin{vmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{vmatrix} \neq 0$.

Example 1. Find the Wronskian of the following pair of functions

$$\begin{aligned} & (i) \text{ cost, sint} \quad (ii) e^{-2t}, te^{-2t} \\ & (iii) x, xe^x \quad (iv) e^t \text{ sint}, e^t \text{ cost.} \end{aligned}$$

Solution

Let $y_1 = e^t \text{ sint}$, $y_2 = e^t \text{ cost}$.

Therefore, $y_1' = e^t \text{ cost} + e^t \text{ sint}$, $y_2' = -e^t \text{ sint} + e^t \text{ cost}$.

Then,

$$\begin{aligned} W &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^t \text{ sint} & e^t \text{ cost} \\ e^t \text{ cost} + e^t \text{ sint} & -e^t \text{ sint} + e^t \text{ cost} \end{vmatrix} \\ &= e^{2t} \begin{vmatrix} \text{sint} & \text{cost} \\ \text{cost} + \text{sint} & -\text{sint} + \text{cost} \end{vmatrix} \\ &= e^{2t} [\text{sint}(-\text{sint} + \text{cost}) - \text{cost}(\text{cost} + \text{sint})] \end{aligned}$$

$$\begin{aligned}
&= e^{2t}[-\sin^2 t + \sin t \cos t - \cos^2 t - \sin t \cos t] \\
&= e^{2t} \times [-1] = -e^{2t}
\end{aligned}$$

Hence,

$$W = -e^{2t}.$$

Fundamental Set of Solutions

Two solutions y_1 and y_2 are said to constitute a fundamental set of solutions of

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

if its every solution can be written as a linear combination of y_1 and y_2 .

Theorem 4 Let y_1 and y_2 be any two solutions of the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1).$$

Then, the family of solutions

$$y = c_1 y_1(t) + c_2 y_2(t)$$

with arbitrary coefficients c_1 and c_2 includes every solution of the differential equation (1) iff there exists a point t_0 such that

$$W(y_1, y_2)(t_0) \neq 0$$

This theorem says:

(i) $y = c_1 y_1(t) + c_2 y_2(t)$ is the general solution of the equation (1) and (ii) y_1, y_2 are said to form a fundamental set of solutions of (1) iff $W(y_1, y_2)(t) \neq 0$ for all t .

Example 1. If $y_1(t) = e^{\lambda t}$ and $y_2(t) = e^{\mu t}$ are two solutions of a differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0,$$

then show that $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions where $\lambda \neq \mu$.

Solution

The given de is

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

has two solutions

$$y_1(t) = e^{\lambda t} \text{ and } y_2(t) = e^{\mu t} \quad \dots \quad (2)$$

$$y_1'(t) = \lambda e^{\lambda t} \text{ and } y_2'(t) = \mu e^{\mu t}$$

$$\begin{aligned}
\text{Then, } W(y_1, y_2)(t) &= \begin{vmatrix} e^{\lambda t} & e^{\mu t} \\ \lambda e^{\lambda t} & \mu e^{\mu t} \end{vmatrix} \\
&= e^{\lambda t} \cdot e^{\mu t} \begin{vmatrix} 1 & 1 \\ \lambda & \mu \end{vmatrix} = e^{(\lambda+\mu)t}(\mu - \lambda)
\end{aligned}$$

Since $\mu \neq \lambda$, $\mu - \lambda \neq 0$ and $e^{(\lambda+\mu)t} \neq 0$ for all t , we have $W(y_1, y_2)(t) \neq 0$.

Hence, y_1 and y_2 form a fundamental set of solutions.

Example 2. Show that $y_1(t) = t^{\frac{1}{2}}$ and $y_2(t) = t^{-1}$ form a fundamental set of solutions of $2t^2 y'' + 3ty' - y = 0, t > 0$.

Solution

The given differential equation is

$$2t^2 y'' + 3ty' - y = 0, t > 0 \quad \dots \quad (1)$$

Next, given that

$$y_1(t) = t^{\frac{1}{2}} \text{ and } y_2(t) = t^{-1}$$

Differentiating, we get

$$y_1'(t) = \frac{1}{2t^{\frac{3}{2}}}, \quad y_2''(t) = -\frac{1}{4t^{\frac{3}{2}}}$$

Putting these in (1), we get

$$\begin{aligned} 2t^2 \left(-\frac{1}{4t^{\frac{3}{2}}} \right) + 3t \left(\frac{1}{2t^{\frac{3}{2}}} \right) - t^{\frac{1}{2}} &= 0 \\ \text{or,} \quad \left(-\frac{1}{2} t^{\frac{1}{2}} \right) + \left(\frac{3}{2} t^{\frac{1}{2}} \right) - t^{\frac{1}{2}} &= 0 \\ \text{or,} \quad -\frac{3}{2} t^{\frac{1}{2}} + \frac{3}{2} t^{\frac{1}{2}} &= 0 \\ \Rightarrow 0 &= 0 \end{aligned}$$

Therefore, $y_1(t) = t^{\frac{1}{2}}$ is a solution of (1).

Also, differentiating $y_2(t) = t^{-1}$, we get

$$y_2'(t) = -t^{-2}, \quad y_2''(t) = 2t^{-3}$$

Using these in (1), we get

$$\begin{aligned} 2t^2(2t^{-3}) + 3t(-t^{-2}) - t^{-1} &= 0 \\ \text{or,} \quad 4t^{-1} - 3t^{-1} - t^{-1} &= 0 \\ \text{or,} \quad 0 &= 0 \end{aligned}$$

Therefore, $y_2(t) = t^{-1}$ is also a solution of (1).

Now,

$$W(y_1, y_2)(t) = \begin{vmatrix} t^{\frac{1}{2}} & t^{-1} \\ \frac{1}{2}t^{-\frac{1}{2}} & -t^{-2} \end{vmatrix} = -t^{-\frac{3}{2}} - \frac{1}{2}t^{-\frac{3}{2}} = -\frac{3}{2}t^{-\frac{3}{2}} \neq 0.$$

Therefore, $W(y_1, y_2)(t) \neq 0$ for all $t > 0$.

Hence, y_1, y_2 form a fundamental set of solutions of (1).

Theorems Let us consider the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

where $p(t), q(t)$ are continuous functions in some interval I . For any t_0 in I , suppose y_1 is a solution of (1) that also satisfies the initial candidates

$$y(t_0) = 1, y'(t_0) = 0$$

and suppose that y_2 is another solution of (1) that satisfies the initial conditions

$$y(t_0) = 0, y'(t_0) = 1.$$

Then, y_1 and y_2 form a fundamental set of solutions of (1)

Example 1. If $y_1(t) = e^t$ and $y_2(t) = e^{-t}$ are solutions of the differential equation

$$y'' - y = 0 \quad \dots \quad (1).$$

Then, show that y_1 and y_2 form a fundamental set of solutions of (1). Also, find the fundamental set of solutions of (1) using the initial point $t_0 = 0$, as indicated by theorem 5.

Solution

The given differential equation is

$$y'' - y = 0 \quad \dots \quad (1)$$

and

$$y_1(t) = e^t \text{ and } y_2(t) = e^{-t}$$

are any two solutions of (1).

Therefore, $y_1'(t) = e^t$ and $y_2'(t) = -e^{-t}$

So, Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_1' \\ y_1' & y_1'' \end{vmatrix} = \begin{vmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{vmatrix} = e^t \cdot e^{-t} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = e^0 \times (-1 - 1) = -2 \neq 0.$$

Therefore, $W(y_1, y_2)(t) \neq 0$ and so $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (1). However, at $t_0 = 0$

$$\begin{aligned} y_1(t_0) &= y_1(0) = 1, y_1'(t_0) = y_1'(0) = 1 \\ y_2(t_0) &= y_2(0) = 1, y_2'(t_0) = y_2'(0) = -1 \end{aligned}$$

So, $y_1(t)$ and $y_2(t)$ do not satisfy the conditions for fundamental set of solutions of (1) as stated by theorem (5).

To find the fundamental set of solutions, consider $y_3(t)$ be a solution of (1) satisfying

$$y_3(0) = 1 \text{ and } y_3'(0) = 0.$$

Suppose

$$y_3(t) = c_1 e^t + c_2 e^{-t}$$

Therefore,

$$y_3'(t) = c_1 e^t - c_2 e^{-t}$$

Using $y_3(0) = 1$ and $y_3'(0) = 0$, we get

$$c_1 + c_2 = 1 \quad \dots \quad (2)$$

and

$$c_1 - c_2 = 0 \quad \dots \quad (3)$$

Solving equations (2) and (3), we get

$$c_1 = \frac{1}{2} \text{ and } c_2 = \frac{1}{2}$$

Therefore,

$$y_3(t) = \frac{1}{2} e^t + \frac{1}{2} e^{-t} = \cosh t.$$

Next, suppose $y_4(t)$ be another solutions of (1) satisfying

$$y_4(0) = 0 \text{ and } y_4'(0) = 1$$

Let, $y_4(t) = d_1 e^t + d_2 e^{-t}$. Therefore,

$$y_4'(t) = d_1 e^t - d_2 e^{-t}$$

Using $y_4(0) = 0$ and $y_4'(0) = 1$, we get

$$d_1 + d_2 = 0 \quad \dots \quad (4)$$

and

$$d_1 - d_2 = 1 \quad \dots \quad (5)$$

Solving (4) and (5), we get

$$d_1 = \frac{1}{2} \text{ and } d_2 = -\frac{1}{2}.$$

Therefore,

$$y_4(t) = \frac{1}{2} e^t - \frac{1}{2} e^{-t} = \sinh t.$$

Now,

$$y_3(t) = \cosh t \Rightarrow y_3'(t) = \sinh t$$

And

$$y_4(t) = \sinh t \Rightarrow y_4'(t) = \cosh t$$

$$\therefore W(y_3, y_4)(t) = \begin{vmatrix} y_3 & y_4 \\ y_3' & y_4' \end{vmatrix} = \begin{vmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{vmatrix} = \cosh^2 t - \sinh^2 t = 1 \neq 0.$$

Hence, $y_3(t) = \cosh t$ and $y_4(t) = \sinh t$ form a fundamental set of solutions of (1) and the general solution of (1) can be written as

$$y = k_1 \cosh t + k_2 \sinh t,$$

where k_1 and k_2 are arbitrary constants.

Theorem 6 (Abel's Theorem)

Let y_1 and y_2 be solutions of the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

where $p(t)$ and $q(t)$ are continuous on an open interval I . Then, the Wronskian $W(y_1, y_2)(t)$ is given by $W(y_1, y_2)(t) = ce^{-\int p(t)dt}$

where c is an arbitrary constant that depends on y_1 and y_2 but not on t . Further, $W(y_1, y_2)(t)$ either is zero for all t on I (i.e. $c = 0$) or else is never zero on I (i.e. $c \neq 0$).

Proof

The given differential equation is

$$L[y] = y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

and let y_1 and y_2 be two solutions of (1) Then,

$$y_1'' + p(t)y_1' + q(t)y_1 = 0 \quad \dots \quad (2)$$

and

$$y_2'' + p(t)y_2' + q(t)y_2 = 0 \quad \dots \quad (3)$$

Multiplying equation (2) by $-y_2$ and (3) by y_1 and by addition, we get

$$(y_1y_2'' - y_1''y_2) + p(t)(y_1y_2' - y_1'y_2) = 0 \quad \dots \quad (4)$$

And

$$\begin{aligned} W &= W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_1'y_2 \\ \Rightarrow W' &= y_1y_2'' + y_1'y_2' - y_1'y_2' - y_1''y_2 = y_1y_2'' - y_1''y_2 \end{aligned}$$

Putting these values in (4), we get

$$\begin{aligned} (y_1y_2'' - y_1''y_2) + p(t)(y_1y_2' - y_1'y_2) &= 0 \\ \Rightarrow W' + p(t)W &= 0 \\ \Rightarrow \frac{dW}{dt} &= -p(t)W \\ \Rightarrow \frac{dW}{W} &= -p(t)dt \\ \Rightarrow \int \frac{dW}{W} &= -\int p(t)dt \\ \Rightarrow \ln W - \ln c &= -\int p(t)dt \\ \Rightarrow \ln \frac{W}{c} &= -\int p(t)dt \\ \Rightarrow \frac{W}{c} &= e^{-\int p(t)dt} \\ \Rightarrow W &= ce^{-\int p(t)dt} \end{aligned}$$

Hence,

$$W(t) = ce^{-\int p(t)dt}$$

where c is an arbitrary constant depending on y_1 and y_2 but not on t .

Also, $e^{-\int p(t)dt} \neq 0$ for every t .

So, for some $t_0 \in I$, if $W(t_0) = 0$, then $c = 0$. This makes $W(t) = 0$ for all t in I .

If $W(t_0) \neq 0$, then $c \neq 0$ and so $W(t) \neq 0$ for all t in I .

This completes the proof of the theorem.

Example 1. Let $y_1(t) = t^{\frac{1}{2}}$ and $y_2(t) = t^{-1}$ be solutions of the differential equation

$$2t^2y'' + 3ty' - y = 0, t > 0.$$

Find the Wronskian $W(y_1, y_2)(t)$ and also verify it by Abel's formula $W(y_1, y_2)(t) = ce^{-\int p(t)dt}$.

Solution

The given differential equation is

$$2t^2y'' + 3ty' - y = 0, t > 0 \quad \dots \quad (1)$$

and $y_1(t) = t^{\frac{1}{2}}$, $y_2(t) = t^{-1}$ are two solutions of (1).

Therefore,

$$y_1'(t) = \frac{1}{2}t^{-\frac{1}{2}}, \quad y_2'(t) = -t^{-2}.$$

So, the Wrosnkian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t^{\frac{1}{2}} & t^{-1} \\ \frac{1}{2}t^{-\frac{1}{2}} & -t^{-2} \end{vmatrix} = -t^{-\frac{3}{2}} - \frac{1}{2}t^{-\frac{3}{2}} = -\frac{3}{2}t^{-\frac{3}{2}}.$$

$$\therefore W = -\frac{3}{2}t^{-\frac{3}{2}}.$$

Next, equation (1) can also be written as

$$y'' + \frac{3}{2t}y' - \frac{1}{2t^2}y = 0.$$

Comparing it with

$$y'' + p(t)y' + q(t)y = 0,$$

we have

$$p(t) = \frac{3}{2t}$$

And so, by Abel's theorem,

$$W(y_1, y_2)(t) = ce^{-\int \frac{3}{2t} dt} = ce^{-\frac{3}{2} \int \frac{1}{t} dt} = ce^{-\frac{3}{2} \ln t} = ce^{\ln t^{-\frac{3}{2}}} = ct^{-\frac{3}{2}}$$

Therefore, $W(y_1, y_2)(t) = ct^{-\frac{3}{2}}$.

Choosing $c = -\frac{3}{2}$, we have

$$W(y_1, y_2)(t) = -\frac{3}{2}t^{-\frac{3}{2}}.$$

Hence, Abels theorem is verified.

Complex Roots of the Auxiliary Equation

The second order linear homogeneous differential equation of constant coefficients is

$$ay'' + 3y' + cy = 0 \quad \dots \quad (1)$$

Let $y = e^{mt}$ be a solution of (1).

Then, the A.E. is

$$am^2 + bm + c = 0 \quad \dots \quad (2)$$

If $b^2 - 4ac < 0$, then the roots (2) are imaginary i.e. m_1 and m_2 are of the form $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where $\beta \neq 0$ and α, β are real numbers. Then,

$$y_1(t) = e^{(\alpha+i\beta)t}, y_2(t) = e^{(\alpha-i\beta)t}$$

are two solutions of (1)

We know, (by Ealer's formula)

$$e^{i\theta} = \cos\theta + i\sin\theta \text{ and } e^{-i\theta} = \cos\theta - i\sin\theta,$$

Therefore,

$$e^{i\beta} = \cos\beta + i\sin\beta \text{ and } e^{-i\beta} = \cos\beta - i\sin\beta$$

Thus,

$$y_1(t) = e^{(\alpha+i\beta)t} = e^{\alpha t} e^{i\beta t} = e^{\alpha t} (\cos\beta t + i\sin\beta t) \quad \dots \quad (2)$$

and

$$y_2(t) = e^{(\alpha-i\beta)t} = e^{\alpha t} e^{-i\beta t} = e^{\alpha t} (\cos\beta t - i\sin\beta t) \quad \dots \quad (3)$$

are two solutions of (1).

Also,

$$y_1'(t) = (\alpha + i\beta)e^{(\alpha+i\beta)t}, \quad y_2'(t) = (\alpha - i\beta)e^{(\alpha-i\beta)t}$$

Then,

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{(\alpha+i\beta)t} & e^{(\alpha-i\beta)t} \\ (\alpha + i\beta)e^{(\alpha+i\beta)t} & (\alpha - i\beta)e^{(\alpha-i\beta)t} \end{vmatrix} \\ &= e^{(\alpha+i\beta)t} \cdot e^{(\alpha-i\beta)t} \begin{vmatrix} 1 & 1 \\ (\alpha + i\beta) & (\alpha - i\beta) \end{vmatrix} \\ &= e^{2\alpha t} (\alpha - i\beta - \alpha - i\beta) = -2i\beta e^{2\alpha t} \neq 0. \end{aligned}$$

Hence, $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (1) and hence, the general solution of (1) can be expressed as a linear combination of $y_1(t)$ and $y_2(t)$ with arbitrary coefficients.

But we are looking for real valued functions of t .

Since, any linear combination of two solution of also a solution,

$$\begin{aligned} y_1(t) + y_2(t) &= e^{\alpha t} (\cos\beta t + i\sin\beta t) + e^{\alpha t} (\cos\beta t - i\sin\beta t) \\ &= 2e^{\alpha t} \cos\beta t \quad \dots \quad (4) \end{aligned}$$

and

$$\begin{aligned} y_1(t) - y_2(t) &= e^{\alpha t} (\cos\beta t + i\sin\beta t) - e^{\alpha t} (\cos\beta t - i\sin\beta t) \\ &= 2ie^{\alpha t} \sin\beta t \quad \dots \quad (5) \end{aligned}$$

are also two solution of (1)

Leaving out the constant 2 and $2i$, we are left with

$$u(t) = e^{\alpha t} \cos\beta t$$

and

$$v(t) = e^{\alpha t} \sin\beta t$$

as the real valued solution of (1)

Also,

$$u'(t) = \alpha e^{\alpha t} \cos\beta t - \beta e^{\alpha t} \sin\beta t$$

and

$$v'(t) = \alpha e^{\alpha t} \sin\beta t + \beta e^{\alpha t} \cos\beta t.$$

$$\begin{aligned} W(u, v)(t) &= \begin{vmatrix} u & v \\ u' & v' \end{vmatrix} \\ &= \begin{vmatrix} e^{\alpha t} \cos\beta t & e^{\alpha t} \sin\beta t \\ \alpha e^{\alpha t} \cos\beta t - \beta e^{\alpha t} \sin\beta t & \alpha e^{\alpha t} \sin\beta t + \beta e^{\alpha t} \cos\beta t \end{vmatrix} = \mu e^{2\alpha t} \end{aligned}$$

where μ is the value of the determinant.

$$\therefore W(u, v)(t) = \mu e^{2\alpha t} \neq 0,$$

where $\mu \neq 0$, otherwise the roots will be real.

Therefore, $u(t)$ and $v(t)$ form a fundamental set of solutions and hence, the general solution of given equation (1) is

$$\begin{aligned} y &= c_1 u(t) + c_2 v(t) \\ \Rightarrow y &= c_1 e^{\alpha t} \cos\beta t + c_2 e^{\alpha t} \sin\beta t \\ \Rightarrow y &= e^{\alpha t} (c_1 \cos\beta t + c_2 \sin\beta t) \end{aligned}$$

Example 1. Find the general solution of the differential equation $y'' + y' + 9.25y = 0$, by finding out the fundamental set of solutions.

Also, find the solution that satisfies the initial conditions

$$y(0) = 2, y'(0) = 8.$$

Solution

The given differential equation is

$$y'' + y' + 9.25y = 0 \quad \dots \quad (1)$$

It's A.E. is

$$m^2 + m + 9.25 = 0$$

Therefore,

$$m = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times 9.25}}{2 \times 1}$$

$$\text{or, } m = \frac{-1 \pm \sqrt{1 - 37}}{2} = \frac{-1 \pm \sqrt{-36}}{2} = \frac{-1 \pm 6i}{2}$$

Hence,

$$m = -\frac{1}{2} + 3i, -\frac{1}{2} - 3i$$

Therefore,

$$y_1(t) = e^{-\frac{t}{2}}(\cos 3t + i \sin 3t) = e^{(-\frac{1}{2} + 3i)t} \quad \dots \quad (2)$$

$$y_2(t) = e^{-\frac{t}{2}}(\cos 3t - i \sin 3t) = e^{(-\frac{1}{2} - 3i)t} \quad \dots \quad (3)$$

Differentiating, we get

$$y_1'(t) = \left(-\frac{1}{2} + 3i\right) e^{(-\frac{1}{2} + 3i)t}$$

And

$$y_2'(t) = \left(-\frac{1}{2} - 3i\right) e^{(-\frac{1}{2} - 3i)t}$$

Then,

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{(-\frac{1}{2} + 3i)t} & e^{(-\frac{1}{2} - 3i)t} \\ \left(-\frac{1}{2} + 3i\right) e^{(-\frac{1}{2} + 3i)t} & \left(-\frac{1}{2} - 3i\right) e^{(-\frac{1}{2} - 3i)t} \end{vmatrix}$$

$$= e^{(-\frac{1}{2} + 3i)t} \times e^{(-\frac{1}{2} - 3i)t} \begin{vmatrix} 1 & 1 \\ -\frac{1}{2} + 3i & -\frac{1}{2} - 3i \end{vmatrix}$$

$$= e^{-t} \left[-\frac{1}{2} - 3i + \frac{1}{2} - 3i \right] = -6ie^{-t} \neq 0.$$

Hence, the general solution of (1) can be expressed as a linear combination of $y_1(t)$ and $y_2(t)$ with arbitrary coefficients. But we need real valued solutions of (1) instead of complex valued solutions $y_1(t)$ and $y_2(t)$. Since, linear combination of two solutions is also a solution,

$$y_1(t) + y_2(t) = 2e^{-\frac{t}{2}} \cos 3t$$

and

$$y_1(t) - y_2(t) = 2ie^{-\frac{t}{2}} \sin 3t.$$

Discarding the constant multipliers 2 and $2i$, we are left with

$$u(t) = e^{-\frac{t}{2}} \cos 3t, v(t) = e^{-\frac{t}{2}} \sin 3t$$

as real valued solution of (1)

Also,

$$u'(t) = -\frac{1}{2} e^{-\frac{t}{2}} \cos 3t - 3e^{-\frac{t}{2}} \sin 3t,$$

and

$$v'(t) = -\frac{1}{2} e^{-\frac{t}{2}} \sin 3t + 3e^{-\frac{t}{2}} \cos 3t$$

Then,

$$\begin{aligned}
W(u, v)(t) &= \begin{vmatrix} u & v \\ u' & v' \end{vmatrix} \\
&= \begin{vmatrix} e^{-\frac{t}{2}} \cos 3t & e^{-\frac{t}{2}} \sin 3t \\ -\frac{1}{2} e^{-\frac{t}{2}} \cos 3t - 3e^{-\frac{t}{2}} \sin 3t & -\frac{1}{2} e^{-\frac{t}{2}} \sin 3t + 3e^{-\frac{t}{2}} \cos 3t \end{vmatrix} \\
&= e^{-\frac{t}{2}} \times e^{-\frac{t}{2}} \begin{vmatrix} \cos 3t & \sin 3t \\ -\frac{1}{2} \cos 3t - 3 \sin 3t & -\frac{1}{2} \sin 3t + 3 \cos 3t \end{vmatrix} \\
&= e^{-t} \left[-\frac{1}{2} \cos 3t \sin 3t + 3 \cos^2 3t + \frac{1}{2} \cos 3t \sin 3t + 3 \sin^2 3t \right] \\
&= e^{-t} [3 \cos^2 3t + 3 \sin^2 3t] \\
&= 3e^{-t}
\end{aligned}$$

$$\therefore W(u, v)(t) = 3e^{-t} \neq 0$$

Hence, $u(t)$ and $v(t)$ form a fundamental set of solutions and the general solution of (1) is

$$y = c_1 u(t) + c_2 v(t) = e^{-\frac{t}{2}} (c_1 \cos 3t + c_2 \sin 3t) \quad \dots \quad (4)$$

where c_1 and c_2 are arbitrary constants.

Next, since $y(0) = 2$, (4) gives

$$2 = 1(c_1 \times 1 + c_2 \times 0) \Rightarrow c_1 = 2$$

Also,

$$y' = -\frac{1}{2} e^{-\frac{t}{2}} (c_1 \cos 3t + c_2 \sin 3t) + e^{-\frac{t}{2}} (-3c_1 \sin 3t + 3c_2 \cos 3t)$$

Again, $y'(0) = 8$ gives

$$\begin{aligned}
8 &= -\frac{1}{2} e^0 (c_1 \times 1 + 0) + e^0 (0 + 3c_2 \times 1) = -\frac{c_1}{2} + 3c_2 \\
&\Rightarrow 8 = -1 + 3c_2 \\
&\Rightarrow 9 = 3c_2 \Rightarrow c_2 = 3
\end{aligned}$$

Hence, required solutions of the initial value problem is

$$y = e^{-\frac{t}{2}} (2 \cos 3t + 3 \sin 3t).$$

Example 2. Find the solution of the IVP

$$y'' + y' + 1.25y = 0, y(0) = 4, y'(0) = 1.$$

Euler Equation

An equation of the form

$$t^2 \frac{d^2 y}{dt^2} + \alpha t \frac{dy}{dt} + \beta y = 0, t > 0 \quad \dots \quad (1)$$

where α, β are real constants, is called an Euler Equation.

Solution of Euler's Equation

The Euler's equation is

$$t^2 \frac{d^2 y}{dt^2} + \alpha t \frac{dy}{dt} + \beta y = 0, t > 0 \quad \dots \quad (1)$$

where α, β are constants.

Let $t = e^x$ i.e. $x = \ln t$.

$$\therefore \frac{dx}{dt} = \frac{1}{t}$$

So,

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \frac{dy}{dx} \times \frac{1}{t} = \frac{1}{t} \frac{dy}{dx}.$$

$$\therefore t \frac{dy}{dt} = \frac{dy}{dx} \quad \dots \quad (2)$$

Also,

$$\begin{aligned} \frac{d^2y}{dt^2} &= \frac{d}{dt} \left(\frac{dy}{dt} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \times \frac{dx}{dt} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \times \frac{1}{t} \right) = -\frac{1}{t^2} \frac{dy}{dx} + \frac{1}{t} \frac{d}{dt} \left(\frac{dy}{dx} \right) \\ &= -\frac{1}{t^2} \frac{dy}{dx} + \frac{1}{t} \left(\frac{d^2y}{dx^2} \times \frac{dx}{dt} \right) = -\frac{1}{t^2} \frac{dy}{dx} + \frac{1}{t} \left(\frac{d^2y}{dx^2} \times \frac{1}{t} \right) \\ &= -\frac{1}{t^2} \frac{dy}{dx} + \frac{1}{t^2} \left(\frac{d^2y}{dx^2} \right) \\ \therefore \frac{d^2y}{dt^2} &= -\frac{1}{t^2} \frac{dy}{dx} + \frac{1}{t^2} \left(\frac{d^2y}{dx^2} \right) \\ \text{or, } t^2 \frac{d^2y}{dt^2} &= \frac{d^2y}{dx^2} - \frac{dy}{dx} \quad \dots \quad (3) \end{aligned}$$

Then, (1) becomes, using (2) and (3).

$$\begin{aligned} \frac{d^2y}{dx^2} - \frac{dy}{dx} + \alpha \frac{dy}{dx} + \beta y &= 0 \\ \text{or, } \frac{d^2y}{dx^2} + (\alpha - 1) \frac{dy}{dx} + \beta y &= 0 \end{aligned}$$

which is a second order homogenous equation with constant coefficients and can be solved for y in terms of x . Later, using $x = \ln t$ gives the required solution.

Examples Solve the following equation.

- i. $t^2 y'' + ty' + y = 0$ for $t > 0$
- ii. $t^2 y'' + 4ty' + 2y = 0, t > 0$

(ii) Solution

The given differential equation is

$$t^2 y'' + 4ty' + 2y = 0, t > 0 \quad \dots \quad (1)$$

Put, $t = e^x$ i.e. $x = \ln t$. Then,

$$\frac{dx}{dt} = \frac{1}{t} \Rightarrow \frac{dx}{dy} \times \frac{dy}{dt} = \frac{1}{t} \Rightarrow t \frac{dy}{dt} = \frac{dy}{dx}$$

And

$$t^2 \frac{d^2y}{dt^2} = \frac{d^2y}{dx^2} - \frac{dy}{dx}$$

Then, equation (1) becomes,

$$\begin{aligned} \frac{d^2y}{dx^2} - \frac{dy}{dx} + 4 \frac{dy}{dx} + 2y &= 0 \\ \text{or, } \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y &= 0 \end{aligned}$$

It's A.E. is

$$\begin{aligned} m^2 + 3m + 2 &= 0 \\ \text{or, } (m + 2)(m + 1) &= 0 \\ \therefore m &= -2, -1 \end{aligned}$$

Then, the general solution is

$$\begin{aligned} y &= c_1 e^{-2x} + c_2 e^{-x} \\ \text{or, } y &= c_1 e^{-2 \ln t} + c_2 e^{-\ln t} \\ \text{or, } y &= c_1 e^{\ln t^{-2}} + c_2 e^{\ln t^{-1}} \end{aligned}$$

Repeated Roots: Reduction of Order

The second order homogeneous differential equation with constant coefficients is

$$ay'' + by' + cy = 0 \quad \dots \quad (1)$$

Let $y = e^{mt}$ be a solution of (1). Then, it's A.E. is

$$am^2 + bm + c = 0 \quad \dots \quad (2)$$

The roots of (2) will be repeated if

$$b^2 - 4ac = 0.$$

And if $b^2 - 4ac = 0$, the repeated root is given by

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b}{2a}$$

$$\therefore m_1 = m_2 = \frac{-b}{2a}.$$

Therefore, $y_1(t) = e^{-\frac{b}{2a}t}$ is one of the solutions of (1).

To find the general solution of (1), we need a second independent solution.

Since $y_1(t)$ is a solution of (1), $cy_1(t)$ is also a solution of (1) where c is an arbitrary constant.

Consider

$$y = v(t).y_1(t) = v(t)e^{-\frac{b}{2a}t} \quad \dots \quad (3)$$

Then,

$$y' = v'(t)e^{-\frac{b}{2a}t} - \frac{b}{2a}v(t)e^{-\frac{b}{2a}t} \quad \dots \quad (4)$$

And

$$y'' = v''(t)e^{-\frac{b}{2a}t} - \frac{b}{2a}v'(t)e^{-\frac{b}{2a}t} - \frac{b}{2a}v'(t)e^{-\frac{b}{2a}t} + \frac{b^2}{4a^2}v(t)e^{-\frac{b}{2a}t}$$

$$\text{or, } y'' = v''(t)e^{-\frac{b}{2a}t} - \frac{b}{a}v'(t)e^{-\frac{b}{2a}t} + \frac{b^2}{4a^2}v(t)e^{-\frac{b}{2a}t} \quad \dots \quad (5)$$

Now, substituting (3), (4) and (5) in (1), we get

$$a \left[v''(t)e^{-\frac{b}{2a}t} - \frac{b}{a}v'(t)e^{-\frac{b}{2a}t} + \frac{b^2}{4a^2}v(t)e^{-\frac{b}{2a}t} \right] + b \left[v'(t)e^{-\frac{b}{2a}t} - \frac{b}{2a}v(t)e^{-\frac{b}{2a}t} \right] + cv(t)e^{-\frac{b}{2a}t} = 0$$

$$\text{or, } \left[av''(t) - bv'(t) + \frac{b^2}{4a}v(t) + bv'(t) - \frac{b^2}{2a}v(t) + cv(t) \right] e^{-\frac{b}{2a}t} = 0$$

$$\text{or, } \left[av''(t) + \left\{ \frac{b^2}{4a} - \frac{b^2}{2a} + c \right\} v(t) \right] e^{-\frac{b}{2a}t} = 0$$

$$\text{or, } \left[av''(t) + \left\{ \frac{b^2 - 2b^2 + 4ac}{4a} \right\} v(t) \right] e^{-\frac{b}{2a}t} = 0$$

$$\text{or, } \left[av''(t) - \left\{ \frac{b^2 - 4ac}{4a} \right\} v(t) \right] e^{-\frac{b}{2a}t} = 0$$

$$\text{or, } av''(t)e^{-\frac{b}{2a}t} = 0, \text{ since } b^2 - 4ac = 0$$

Again, since $e^{-\frac{b}{2a}t} \neq 0, a \neq 0$, we have

$$v''(t) = 0.$$

Integrating, we have

$$v'(t) = c_1.$$

Again, integrating, we have

$$v(t) = c_1t + c_2$$

where c_1 and c_2 are arbitrary constants.

From (3)

$$y = (c_1 t + c_2) e^{-\frac{b}{2a}t}$$

$$\text{or, } y = \left(c_1 t e^{-\frac{b}{2a}t} + c_2 e^{-\frac{b}{2a}t} \right)$$

That is, y is a linear combination of two solutions $y_1(t) = t e^{-\frac{b}{2a}t}$ and $y_2(t) = e^{-\frac{b}{2a}t}$.
Now,

$$y_1'(t) = e^{-\frac{b}{2a}t} - \frac{b}{2a} t e^{-\frac{b}{2a}t} = \left(1 - \frac{bt}{2a} \right) e^{-\frac{b}{2a}t}$$

And

$$y_2'(t) = -\frac{b}{2a} e^{-\frac{b}{2a}t}.$$

So,

$$\begin{aligned} W(y_1, y_2) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t e^{-\frac{b}{2a}t} & e^{-\frac{b}{2a}t} \\ \left(1 - \frac{bt}{2a} \right) e^{-\frac{b}{2a}t} & -\frac{b}{2a} e^{-\frac{b}{2a}t} \end{vmatrix} \\ &= e^{-\frac{b}{2a}t} \times e^{-\frac{b}{2a}t} \begin{vmatrix} t & 1 \\ \left(1 - \frac{bt}{2a} \right) & -\frac{b}{2a} \end{vmatrix} \\ &= e^{-\frac{bt}{a}} \left\{ -\frac{bt}{2a} - 1 + \frac{bt}{2a} \right\} = -e^{-\frac{bt}{a}} \neq 0, \forall t \in \mathbb{R}. \end{aligned}$$

Therefore, the solutions $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (1).
Hence, the general solution of (1) is

$$y = \left(c_1 t e^{-\frac{b}{2a}t} + c_2 e^{-\frac{b}{2a}t} \right) = (c_1 t + c_2) e^{-\frac{b}{2a}t}.$$

Reduction of Order

Let us consider $y_1(t)$ as one of the solutions of

$$y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

To get second independent solution we assume

$$y = v(t)y_1(t) \quad \dots \quad (2)$$

Then,

$$y' = v'(t)y_1(t) + v(t)y_1'(t) \quad \dots \quad (3)$$

and

$$\begin{aligned} y'' &= v''(t)y_1(t) + v'(t)y_1'(t) + v'(t)y_1'(t) + v(t)y_1''(t) \\ \text{or, } y'' &= v''(t)y_1(t) + 2v'(t)y_1'(t) + v(t)y_1''(t) \quad \dots \quad (4) \end{aligned}$$

Putting (2), (3) and (4) in (1), we get

$$\begin{aligned} y_1 v'' + 2y_1' v' + y_1'' v + p(y_1 v' + y_1' v) + q y_1 v &= 0 \\ \text{or, } y_1 v'' + (2y_1' + p y_1) v' + (y_1'' + p y_1' + q y_1) v &= 0 \quad \dots \quad (5) \end{aligned}$$

Since y_1 is a solution of (1)

$$y_1'' + p y_1' + q y_1 = 0.$$

Then, equation (5) reduces to

$$y_1 v'' + (2y_1' + p y_1) v' = 0 \quad \dots \quad (6)$$

which is a first order differential equation for the function v' and can be solved either as a first order linear differential equation or as a separable equation.

After finding v' , v is obtained by further integration. Finally, y is obtained from (2).

This method is called the method of reduction of order.

Example 1. Solve the differential equation $y'' + 4y' + 4y = 0$ using reduction of order.

Solution

The given DE is

$$y'' + 4y' + 4y = 0 \quad \dots \quad (1)$$

Let $y = e^{mt}$ be a solution of (1). Then, it's A.E. is

$$m^2 + 4m + 4 = 0 \quad \dots \quad (2)$$

Here, $b^2 - 4ac = 4^2 - 4.1.4 = 0$. Therefore, the roots of (2) are equal.

$$i.e. m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = -\frac{b}{2a} = -\frac{4}{2} = -2.$$

That is, $m_1 = m_2 = -2$.

Therefore, $y_1 = e^{-2t}$ is one of the solutions of (1).

To find the general solution of (1), we need a second independent solution. Since

$$y_1 = e^{-2t}$$

is a solution of (1),

$$cy_1 = ce^{-2t}$$

is also a solution of (1) where c is an arbitrary constant.

Let us replace c by a function $v(t)$. Consider

$$y = v(t)y_1(t) = v(t)e^{-2t} \quad \dots \quad (3)$$

$$\Rightarrow y' = v'(t)e^{-2t} - 2v(t)e^{-2t} \quad \dots \quad (4)$$

And

$$y'' = v''(t)e^{-2t} - 2v'(t)e^{-2t} - 2v'(t)e^{-2t} + 4v(t)e^{-2t}$$

$$y'' = v''(t)e^{-2t} - 4v'(t)e^{-2t} + 4v(t)e^{-2t} \quad \dots \quad (5)$$

Substituting these values from (3), (4) and (5) in (1), we get

$$\begin{aligned} y'' + 4y' + 4y &= 0 \\ v''(t)e^{-2t} - 4v'(t)e^{-2t} + 4v(t)e^{-2t} + 4v'(t)e^{-2t} - 8v(t)e^{-2t} + 4v(t)e^{-2t} &= 0 \\ \text{or, } v''(t)e^{-2t} &= 0 \end{aligned}$$

Since $e^{-2t} \neq 0$, we must have $v''(t) = 0$.

Integration implies

$$v'(t) = c_2$$

Again, integrating, we get

$$v(t) = c_2t + c_1$$

Therefore, from (3), we get

$$y = c_2te^{-2t} + c_1e^{-2t} = (c_1 + c_2t)e^{-2t}$$

Hence y is a linear combination of the two solutions:

$$y_1(t) = e^{-2t} \text{ and } y_2(t) = te^{-2t}.$$

Therefore,

$$y_1'(t) = -2e^{-2t} \text{ and } y_2'(t) = e^{-2t} - 2te^{-2t}$$

Now,

$$\begin{aligned} W(y_1, y_2) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-2t} & te^{-2t} \\ -2e^{-2t} & e^{-2t} - 2te^{-2t} \end{vmatrix} \\ &= e^{-2t} \times e^{-2t} \begin{vmatrix} 1 & t \\ -2 & 1 - 2t \end{vmatrix} \\ &= e^{-4t} \{1 - 2t + 2t\} = e^{-4t} \neq 0 \quad \forall t \in R. \end{aligned}$$

Therefore, the solutions $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (1). Hence, the general solution of (1) is

$$y = (c_1 + c_2t)e^{-2t}$$

where c_1 and c_2 are arbitrary constants.

Example 2. Find the solution of the differential equation

$$t^2 y'' + 2ty' - 2y = 0, t > 0, y_1(t) = t$$

using the method of reduction of order.

Solution

The given differential equation is

$$t^2 y'' + 2ty' - 2y = 0, t > 0 \quad \dots \quad (1)$$

Given $y_1(t) = t$ is a solution of (1), since

$$y_1'(t) = 1, y_1''(t) = 0$$

satisfy (1).

To find the second fundamental solution of (1), we consider

$$y = v(t)y_1(t) = v(t).t \quad \dots \quad (2)$$

Therefore,

$$y' = v'(t).t + v(t) \quad \dots \quad (3)$$

And

$$y'' = v''(t).t + v'(t) + v'(t) = v''(t).t + 2v'(t) \quad \dots \quad (4)$$

Using (2), (3) and (4) in (1) we get

$$\begin{aligned} t^2\{v''(t).t + 2v'(t)\} + 2t\{v'(t).t + v(t)\} - 2v(t).t &= 0 \\ \text{or, } t^3 v''(t) + 2t^2 v'(t) + 2t^2 v'(t) + 2tv(t) - 2tv(t) &= 0 \\ \text{or, } t^3 v''(t) + 4t^2 v'(t) &= 0 \quad \dots \quad (5) \end{aligned}$$

Since t is a solution of (1), equation (5) becomes

$$\begin{aligned} t^3 v''(t) &= -4t^2 v'(t) \\ \text{or, } tv''(t) &= -4v'(t) \\ \text{or, } \frac{dv'(t)}{dt} &= -4 \frac{v'(t)}{t} \\ \text{or, } \frac{dv'(t)}{v'(t)} &= -4 \frac{dt}{t} \\ \text{or, } \ln v'(t) &= -4 \ln t + \ln c_1 \\ \text{or, } \ln v'(t) &= \ln t^{-4} + \ln c_1 \\ \text{or, } \ln v'(t) &= \ln(c_1 t^{-4}) \\ \text{or, } v'(t) &= c_1 t^{-4} \end{aligned}$$

Again, integrating we get

$$v(t) = \frac{-c_1}{3} t^{-3} + c_2$$

Putting this value of $v(t)$ in (2), we get

$$\begin{aligned} y &= \left(\frac{-c_1}{3} t^{-3} + c_2 \right).t \\ y &= -\frac{c_1}{3} t^{-2} + c_2 t \end{aligned}$$

Hence, y is a linear combination of two solutions

$$y_1(t) = t \text{ and } y_2(t) = t^{-2}.$$

Then,

$$y_1'(t) = 1 \text{ and } y_2'(t) = -2t^{-3}$$

So,

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t & t^{-2} \\ 1 & -2t^{-3} \end{vmatrix} \\ &= -2t^{-2} - t^{-2} = -3t^{-2} \neq 0 \text{ for all } t. \end{aligned}$$

Hence, $y_1(t) = t$ and $y_2(t) = t^{-2}$ form a fundamental set of solutions of (1). So, the general solution is

$$y = d_1 t + d_2 t^{-2}.$$

Non-homogeneous Equations, Method of Undetermined Coefficients

The second order linear equation of the form

$$L[y] = y'' + p(t)y' + q(t)y = g(t) \quad \dots \quad (1)$$

where $p(t)$, $q(t)$ and $g(t)$ are continuous functions on an open interval I and $g(t) \neq 0$, is called a non-homogeneous equation. If $g(t) = 0$, then equation (1) becomes

$$L[y] = y'' + p(t)y' + q(t)y = 0$$

which is called the homogeneous equation corresponding to (1).

Theorem 1 If Y_1 and Y_2 are two solutions of the non-homogeneous equation

$L[y] = y'' + p(t)y' + q(t)y = g(t)$, then their difference $Y_1 - Y_2$ is a solution of the corresponding homogeneous equation $L[y] = 0$. If y_1, y_2 form a fundamental set of solutions of $L[y] = 0$, then $Y_1(t) - Y_2(t) = c_1 y_1(t) - c_2 y_2(t)$, where c_1 and c_2 are arbitrary constants.

Proof

The given non-homogeneous DE is

$$L[y] = g(t) \quad \dots \quad (1)$$

where $L[y] = y'' + p(t)y' + q(t)y$.

Since Y_1 , and Y_2 are solutions of (1), we have

$$L[Y_1] = g(t) \text{ and } L[Y_2] = g(t).$$

Then,

$$\begin{aligned} L[Y_1] - L[Y_2] &= g(t) - g(t) = 0 \\ \Rightarrow L[Y_1 - Y_2] &= 0, \text{ since } L \text{ is linear.} \end{aligned}$$

Therefore, $Y_1 - Y_2$ is a solution of the homogeneous equation $L[y] = 0$. Also, if y_1 and y_2 form a fundamental set of solutions of $L[y] = 0$ and $Y_1 - Y_2$ is a solution of $L[y] = 0$, then $Y_1 - Y_2$ can be expressed as a linear combination of y_1 and y_2 that is,

$$Y_1 - Y_2 = c_1 y_1 + c_2 y_2$$

where c_1 and c_2 are arbitrary constants.

This completes the proof of the theorem.

Theorem 2

If y_1 and y_2 form a fundamental set of solutions of $L[y] = 0$, then the general solution of $L[y] = g(t)$ can be written as

$$y = \phi(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t)$$

where c_1 and c_2 are arbitrary constants and Y is some specific solution of $L[y] = g(t)$.

Proof

We first show that

$$y = \phi(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t) \quad \dots \quad (1)$$

is a solution of

$$L[y] = g(t) \quad \dots \quad (2)$$

Where c_1 and c_2 are arbitrary constants and Y is some specific solution of (2) and y_1, y_2 are fundamental solutions of $L[y] = 0$

Differentiating (1) we get

$$\begin{aligned} y' &= c_1 y_1' + c_2 y_2' + Y' \\ y'' &= c_1 y_1'' + c_2 y_2'' + Y'' \end{aligned}$$

$$\begin{aligned}
& \therefore y'' + p(t)y' + q(t)y \\
&= (c_1y_1'' + c_2y_2'' + Y'') + p(t)(c_1y_1' + c_2y_2' + Y') + q(t)(c_1y_1 + c_2y_2 + Y) \\
&= c_1\{y_1'' + p(t)y_1' + q(t)y_1\} + c_2\{y_2'' + p(t)y_2' + q(t)y_2\} + \{Y'' + p(t)Y' + q(t)Y\} \\
&= c_1 \cdot 0 + c_2 \cdot 0 + g(t) = g(t).
\end{aligned}$$

Therefore, (1) is a solution of

$$L[y] \equiv y'' + p(t)y' + q(t)y = g(t).$$

Next, we show that every solution of (2) is in the form of (1) for some choice of the constants c_1 and c_2 . Let y be any solution of (2). We claim that $y - Y$ is a solution of the homogeneous equation $L[y] = 0$.

That is,

$$\begin{aligned}
L[y - Y] &= (y - Y)'' + p(t)(y - Y)' + q(t)(y - Y) \\
&= (y'' - Y'') + p(t)(y' - Y') + q(t)(y - Y) \\
&= [y'' + p(t)y' + q(t)y] - [Y'' + p(t)Y' + q(t)Y] \\
&= g(t) - g(t) = 0
\end{aligned}$$

This implies that $y - Y$ is a solution of the homogeneous equation $L[y] = 0$ with y_1 and y_2 as fundamental solutions. Then, by theorem 1

$$\begin{aligned}
y(t) - Y(t) &= c_1y_1(t) + c_2y_2(t) \\
\Rightarrow y(t) &= c_1y_1(t) + c_2y_2(t) + Y(t)
\end{aligned}$$

for some constants c_1 and c_2 . This completes the proof.

Note

The general solution of the DE $L[y] = g(t)$ contains two parts $c_1y_1(t) + c_2y_2(t)$ and $Y(t)$. The first part $c_1y_1(t) + c_2y_2(t)$ is the general solution of the corresponding homogeneous equation $L[y] = 0$. This solution is called complementary solution and is denoted by $y_c(t)$. And the second part $Y(t)$ is a single solution or particular solution of the non-homogeneous equation $L[y] = g(t)$. It is denoted by $y_p(t)$ or $Y(t)$.

Hence, the general solution of $L[y] = g(t)$ is

$$y = y_c(t) + Y(t).$$

Method of Undetermined Coefficients

I. When the RHS is an Exponential Function $g(t) = e^{at}$ for $L[G] \neq 0$.

The second order non-homogeneous ODE is

$$\begin{aligned}
L[y] &= y'' + py' + qy = g(t) \text{ where } g(t) = e^{at} \quad \dots \quad (1) \\
\therefore y'' + py' + qy &= e^{at}
\end{aligned}$$

Let $Y(t)$ be some specific solution of (1). Then,

$$Y'' + pY' + qY = e^{at} \quad \dots \quad (2)$$

Case I $L[a] = a^2 + pa + q \neq 0$

Suppose $Y(t) = Ae^{at}$.

$$[\because Y(t) \text{ is some multiple of } e^{at}]$$

Then,

$$Y'(t) = aAe^{at} \text{ and } Y''(t) = a^2Ae^{at}$$

Thus, (2) reduces to

$$\begin{aligned}
a^2Ae^{at} + paAe^{at} + qAe^{at} &= e^{at} \\
\text{or, } A(a^2 + pa + q)e^{at} &= e^{at} \\
\text{or, } A(a^2 + pa + q) &= 1
\end{aligned}$$

$$\text{or, } A = \frac{1}{a^2 + pa + q}$$

Thus, a particular solution (P.S.) is

$$Y(t) = \frac{1}{a^2 + pa + q} e^{at}$$

where $a^2 + pa + q \neq 0$.

Case II When $L[a] = a^2 + pa + q = 0$

Suppose $Y(t) = Ate^{at}$. Then,

$$Y'(t) = Aate^{at} + Ae^{at}$$

$$\text{and } Y''(t) = Aa^2te^{at} + Aae^{at} + Aae^{at} = Aa^2te^{at} + 2Aae^{at}$$

Then (2) reduces to

$$Aa^2te^{at} + 2Aae^{at} + p\{Aate^{at} + Ae^{at}\} + q\{Aae^{at}\} = e^{at}$$

$$(2a + p)Ae^{at} + \{a^2 + ap + q\}Ate^{at} = e^{at}$$

Since $a^2 + ap + q = 0$,

$$(2a + p)Ae^{at} = e^{at}$$

$$\text{or, } (2a + p)A = 1$$

$$\text{or, } A = \frac{1}{2a + p}$$

Hence, the required particular solution is

$$Y(t) = \frac{1}{2a + p} e^{at}.$$

Example 1. Find a particular solution of $y'' - 3y' - 4y = 3e^{2t}$.

Solution

The given equation is

$$y'' - 3y' - 4y = 3e^{2t} \quad \dots \quad (1)$$

Let $Y(t) = Ae^{2t}$ be some specific solution of (1). Then,

$$Y'(t) = 2Ae^{2t}$$

And

$$Y''(t) = 4Ae^{2t}.$$

Thus, (1) reduces to

$$Y'' - 3Y' - 4Y = 3e^{2t}$$

$$\text{or, } 4Ae^{2t} - 3 \times 2Ae^{2t} - 4Ae^{2t} = 3e^{2t}$$

$$\text{or, } 4Ae^{2t} - 6Ae^{2t} - 4Ae^{2t} = 3e^{2t}$$

$$\text{or, } -6A = 3 \Rightarrow A = -\frac{1}{2}.$$

Hence, the required particular solution is

$$Y(t) = -\frac{1}{2}e^{2t}.$$

Example 2. Find particular solution of $y'' - 3y' - 4y = 2e^{-t}$

Solution

The given ODE is

$$y'' - 3y' - 4y = 2e^{-t} \quad \dots \quad (1)$$

Here $a = -1$ and

$$L[a] = L[-1] = (-1)^2 - 3(-1) - 4 = 0$$

So, let $Y(t) = Ate^{-t}$.

Then, $Y'(t) = -Ate^{-t} + Ae^{-t}$

$$\text{and } Y''(t) = Ate^{-t} - Ae^{-t} - Ae^{-t} = Ate^{-t} - 2Ae^{-t}.$$

Then (1) reduces to

$$\begin{aligned} Ate^{-t} - 2Ae^{-t} - 3(-Ate^{-t} + Ae^{-t}) - 4 \times Ate^{-t} &= 2e^{-t} \\ \text{or, } (-2A - 3A)e^{-t} + (A + 3A - 4A)te^{-t} &= 2e^{-t} \\ \text{or, } -5Ae^{-t} &= 2e^{-t} \\ \text{or, } -5A &= 2 \Rightarrow A = -\frac{2}{5}. \end{aligned}$$

Hence, the required particular solution is

$$Y(t) = -\frac{2}{5}te^{-t}.$$

II When the RHS $g(t)$ is a sine function. i.e. $g(t) = \sin at$.

The second order non-homogeneous linear ODE is

$$L[y] = y'' + py' + qy = \sin at \quad \dots \quad (1)$$

Let

$$Y(t) = A \sin at + B \cos at$$

be some specific solution of (1).

Then,

$$Y'(t) = Aa \cos at - Ba \sin at$$

And

$$Y''(t) = -Aa^2 \sin at - Ba^2 \cos at$$

Then, (1) reduces to

$$\begin{aligned} -Aa^2 \sin at - Ba^2 \cos at + p\{Aa \cos at - Ba \sin at\} + q\{A \sin at + B \cos at\} &= \sin at \\ \text{or, } \{-Aa^2 - pBa + qA\} \sin at + \{-Ba^2 + pAa + qB\} \cos at &= \sin at \end{aligned}$$

Equating the coefficients of like terms, we get

$$-Aa^2 - pBa + qA = 1 \Rightarrow (-a^2 + q)A - paB - 1 = 0$$

And

$$-Ba^2 + pAa + qB = 0 \Rightarrow paA + (-a^2 + q)B + 0 = 0$$

Solving by cross multiplication method,

$$\begin{aligned} \frac{A}{-a^2 + q} &= \frac{B}{-ap} = \frac{1}{(-a^2 + q)(-a^2 + q) + a^2p^2} \\ \Rightarrow \frac{A}{-a^2 + q} &= \frac{B}{-ap} = \frac{1}{a^4 - 2a^2q + q^2 + a^2p^2} \\ \therefore A &= \frac{-a^2 + q}{a^4 - 2a^2q + q^2 + a^2p^2}, \quad B = \frac{-ap}{a^4 - 2a^2q + q^2 + a^2p^2} \end{aligned}$$

Hence, the required particular solution is

$$Y(t) = \frac{-a^2 + q}{a^4 - 2a^2q + q^2 + a^2p^2} \sin at - \frac{ap}{a^4 - 2a^2q + q^2 + a^2p^2} \cos at$$

Example 2. Find the particular solution of $y'' - 3y' - 4y = 2 \sin t$.

Solution

The given equation is

$$L[y] = y'' - 3y' - 4y = 2 \sin t \quad \dots \quad (1)$$

Let

$$Y(t) = A \sin t + B \cos t$$

be some specific solution of (1).

Then,

$$Y'(t) = A \cos t - B \sin t$$

and

$$Y''(t) = -Asint - Bcost.$$

Therefore, (1) reduces to

$$\begin{aligned} -Asint - Bcost - 3(Acost - Bsint) - 4(Asint + Bcost) &= 2sint \\ \text{or, } (-A + 3B - 4A)sint + (-B - 3A - 4B)cost &= 2sint \\ \text{or, } (-5A + 3B)sint + (-3A - 5B)cost &= 2sint \end{aligned}$$

Equating the coefficients of like terms, we get

$$-5A + 3B = 2 \text{ and } -3A - 5B = 0$$

Solving, we get

$$A = -\frac{5}{17}, B = \frac{3}{17}.$$

Hence, the required particular solution is

$$Y(t) = -\frac{5}{17}sint + \frac{3}{17}cost.$$

III When the R.H.S. $g(t)$ is a Product of a Trigonometric and an Exponential function, that is, $g(t) = e^{at} \cos bt$.

The second order non-homogeneous linear ODE is

$$L[y] = y'' + py' + qy = e^{at} \cos bt \quad \dots \quad (1)$$

Let

$$Y(t) = Ae^{at} \cos bt + Be^{at} \sin bt \quad \dots \quad (2)$$

be some specific solution of (1).

Then,

$$\begin{aligned} Y'(t) &= A[ae^{at} \cos bt - be^{at} \sin bt] + B\{ae^{at} \sin bt + be^{at} \cos bt\} \\ &= [aA + bB]e^{at} \cos bt + \{-bA + aB\}e^{at} \sin bt \\ Y''(t) &= [(a^2 - b^2)A + 2abB]e^{at} \cos bt + \{-2abA + (a^2 - b^2)B\}e^{at} \sin bt \end{aligned}$$

Then, (1) becomes

$$\begin{aligned} &[(a^2 - b^2)A + 2abB]e^{at} \cos bt + \{-2abA + (a^2 - b^2)B\}e^{at} \sin bt \\ &+ p\{[aA + bB]e^{at} \cos bt + \{-bA + aB\}e^{at} \sin bt\} + q\{Ae^{at} \cos bt + Be^{at} \sin bt\} \\ &= e^{at} \cos bt \\ \text{or, } &[(a^2 - b^2 + ap + q)A + (2ab + pb)B]e^{at} \cos bt \\ &+ \{(-2ab - pb)A + (a^2 - b^2 + ap + q)B\}e^{at} \sin bt = e^{at} \cos bt \end{aligned}$$

Equating the coefficients of like terms, we get

$$(a^2 - b^2 + ap + q)A + (2ab + pb)B = 1$$

And

$$(-2ab - pb)A + (a^2 - b^2 + ap + q)B = 0.$$

Solving these two equations to get A and B which when substituted in (2), we get the required particular solution of (1).

Example 1. Find the particular solution of $y'' - 3y' - 4y = -8e^t \cos 2t$.

Solution

The given equation is

$$y'' - 3y' - 4y = -8e^t \cos 2t \quad \dots \quad (1)$$

Let

$$Y(t) = Ae^t \cos 2t + Be^t \sin 2t \quad \dots \quad (2)$$

be a specific solution of (1)

Then,

$$\begin{aligned} Y'(t) &= Ae^t \cos 2t - 2Ae^t \sin 2t + Be^t \sin 2t + 2Be^t \cos 2t \\ &= A(e^t \cos 2t - 2e^t \sin 2t) + B(e^t \sin 2t + 2e^t \cos 2t) \end{aligned}$$

$$= (A + 2B)e^t \cos 2t + (-2A - B)e^t \sin 2t$$

And

$$\begin{aligned} Y''(t) &= (A + 2B)\{e^t \cos 2t - 2e^t \sin 2t\} + (-2A - B)\{e^t \sin 2t + 2e^t \cos 2t\} \\ &= (-3A + 4B)e^t \cos 2t + (-4A - 3B)e^t \sin 2t \end{aligned}$$

Then, (1) reduces to

$$\begin{aligned} &(-3A + 4B)e^t \cos 2t + (-4A - 3B)e^t \sin 2t - 3[(A + 2B)e^t \cos 2t + (-2A - B)e^t \sin 2t] \\ &\quad - 4\{e^t \cos 2t + Be^t \sin 2t\} = -8e^t \cos 2t \\ &\{-3A + 4B - 3A - 6B - 4A\}e^t \cos 2t + \{-4A - 3B + 6A - 3B - 4B\}e^t \sin 2t = -8e^t \cos 2t \end{aligned}$$

$$\{-10A - 2B\}e^t \cos 2t + \{2A - 10B\}e^t \sin 2t = -8e^t \cos 2t$$

Equating the coefficients of like terms, we get

$$-10A - 2B = -8 \Rightarrow 5A + B = 4$$

And

$$2A - 10B = 0 \Rightarrow A - 5B = 0.$$

Solution gives

$$A = \frac{10}{13}, B = \frac{2}{13}.$$

Hence, the required particular solution is

$$Y(t) = \frac{10}{13}e^t \cos 2t + \frac{2}{13}e^t \sin 2t$$

IV When the RHS is a polynomial, that is,

$$g(t) = a_n t^n + a_{n-1} t^{n-1} + \dots + a_0.$$

Let us consider the second order non-homogeneous ODE

$$ay'' + by' + cy = a_n t^n + a_{n-1} t^{n-1} + \dots + a_0.$$

Let

$$Y(t) = A_n t^n + A_{n-1} t^{n-1} + \dots + A_0 \quad \dots \quad (2)$$

be some specific solution of (1). Then,

$$Y'(t) = nA_n t^{n-1} + (n-1)A_{n-1} t^{n-2} + \dots + A_1$$

And

$$Y''(t) = n(n-1)A_n t^{n-2} + (n-1)(n-2)A_{n-1} t^{n-3} + \dots + A_2$$

Now, equation (1) becomes

$$\begin{aligned} &a[n(n-1)A_n t^{n-2} + (n-1)(n-2)A_{n-1} t^{n-3} + \dots + 2A_2] \\ &\quad + b[nA_n t^{n-1} + (n-1)A_{n-1} t^{n-2} + \dots + A_1] \\ &\quad + c[A_n t^n + A_{n-1} t^{n-1} + \dots + A_0] = a_n t^n + a_{n-1} t^{n-1} + \dots + a_0 \quad \dots \quad (3) \end{aligned}$$

Equating the coefficients of like terms, we get

$$cA_n = a_n \quad \dots \quad (4)$$

$$cA_{n-1} + nbA_n = a_{n-1} \quad \dots \quad (5)$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$cA_0 + bA_1 + 2aA_2 = a_0 \quad \dots \quad (6)$$

Case I $c \neq 0$

If $c \neq 0$, then from (4), $A_n = \frac{a_n}{c}$

and the remaining equations give the values of $A_1, A_2, \dots$, successively.

Case II $c = 0$

If $c = 0$ but $b \neq 0$, then left side of (3) is of degree $n - 1$ and right side is of degree n . In this case, we must choose $Y(t)$ of degree $n + 1$. So, suppose

$$Y(t) = t[A_n t^n + A_{n-1} t^{n-1} + \dots + A_0].$$

Since $b \neq 0$, we can get $A_n = \frac{a_n}{b(n+1)}$ as in case (1) and other coefficients $A_1, A_2, \dots, A_{n-1}$ can be determined in the similar manner.

Case III $c = 0, b = 0$

If $c = 0, b = 0$, then the polynomial of left side of equation (3) is of degree $n - 2$. Similar to case II, we choose

$$Y(t) = t^2[A_n t^n + A_{n-1} t^{n-1} + \dots + A_0]$$

and proceed.

V. When RHS is of the Form $g(t) = e^{\alpha t} P_n(t)$.

Consider the non-homogeneous linear ODE

$$ay'' + by' + cy = e^{\alpha t} P_n(t) \quad \dots \quad (1)$$

Let

$$Y(t) = e^{\alpha t} u(t)$$

be a solution of (1).

Then,

$$Y'(t) = e^{\alpha t} u'(t) + \alpha e^{\alpha t} u(t) = e^{\alpha t} [u'(t) + \alpha u(t)]$$

And

$$Y''(t) = e^{\alpha t} [u''(t) + \alpha u'(t)] + \alpha e^{\alpha t} [u'(t) + \alpha u(t)] = e^{\alpha t} [u''(t) + 2\alpha u'(t) + \alpha^2 u(t)].$$

Then, the equation (1) becomes

$$\begin{aligned} a e^{\alpha t} [u''(t) + 2\alpha u'(t) + \alpha^2 u(t)] + b e^{\alpha t} [u'(t) + \alpha u(t)] + c e^{\alpha t} u(t) &= e^{\alpha t} P_n(t) \\ \Rightarrow a u''(t) + (2a\alpha + b) u'(t) + (a\alpha^2 + b\alpha + c) u(t) &= P_n(t) \end{aligned}$$

Case I $a\alpha^2 + b\alpha + c \neq 0$

If $a\alpha^2 + b\alpha + c \neq 0$, then we consider

$$u(t) = A_n t^n + \dots + A_0$$

and hence,

$$Y(t) = e^{\alpha t} [A_n t^n + \dots + A_0]$$

will be some effeific solution of (1).

Case II $a\alpha^2 + b\alpha + c = 0$ and $2a\alpha + b \neq 0$

If $a\alpha^2 + b\alpha + c = 0$ and $2a\alpha + b \neq 0$, then we consider

$$u(t) = t[A_n t^n + \dots + A_0]$$

and hence

$$Y(t) = e^{\alpha t} t[A_n t^n + \dots + A_0]$$

will be some specific solution of (1).

Case III $a\alpha^2 + b\alpha + c = 0$ and $2a\alpha + b = 0$

If $a\alpha^2 + b\alpha + c = 0$ and $2a\alpha + b = 0$, then we consider

$$u(t) = t^2[A_n t^n + \dots + A_0]$$

and hence,

$$Y(t) = e^{\alpha t} t^2[A_n t^n + \dots + A_0]$$

will be somparticular solution of (1).

VI. When the RHS is of the form $g(t) = e^{\alpha t} P_n(t) \cos \beta t$ or $e^{\alpha t} P_n(t) \sin \beta t$.

Consider the second order non-homogeneous linear ODE

$$ay'' + by' + cy = e^{\alpha t} P_n(t) \sin \beta t$$

By Euler's formula

$$\sin\beta t = \frac{e^{i\beta t} - e^{-i\beta t}}{2i}.$$

Then,

$$\begin{aligned} g(t) &= e^{\alpha t} P_n(t) \sin\beta t = e^{\alpha t} P_n(t) \left(\frac{e^{i\beta t} - e^{-i\beta t}}{2i} \right) \\ &= P_n(t) \left(\frac{e^{(\alpha+i\beta)t} - e^{(\alpha-i\beta)t}}{2i} \right). \end{aligned}$$

In this case we choose a particular solution as

$$Y(t) = e^{(\alpha+i\beta)t} (A_n t^n + A_{n-1} t^{n-1} + \dots + A_0) + e^{(\alpha-i\beta)t} (B_n t^n + B_{n-1} t^{n-1} + \dots + B_0)$$

Or,

$$Y(t) = e^{\alpha t} (A_n t^n + A_{n-1} t^{n-1} + \dots + A_0) \cos\beta t + e^{\alpha t} (B_n t^n + B_{n-1} t^{n-1} + \dots + B_0) \sin\beta t.$$

Example 1. Find the general solution of the given IVP

$$y'' + y' - 2y = 2t, y(0) = 0, y'(0) = 1.$$

Solution

The given equation is

$$y'' + y' - 2y = 2t \quad \dots \quad (1)$$

with initial conditions

$$y(0) = 0, y'(0) = 1$$

It's A.E. equation is

$$\begin{aligned} m^2 + m - 2 &= 0 \\ \text{or, } (m-1)(m+2) &= 0 \\ \therefore m &= 1, -2. \end{aligned}$$

So, the complementary solution of (1) is

$$y_c(t) = c_1 e^t + c_2 e^{-2t} \quad \dots \quad (2)$$

For particular solution,

$$\text{let } Y(t) = At + B.$$

Then,

$$Y'(t) = A \text{ and } Y''(t) = 0.$$

From (1),

$$\begin{aligned} 0 + A - 2At - 2B &= 2t \\ \text{or, } 2At + (A - 2B) &= 2t \end{aligned}$$

Equating the coefficients of like terms, we get

$$-2A = 2, \quad A - 2B = 0$$

Solving, we get $A = -1, B = -\frac{1}{2}$.

Therefore, the particular solution is

$$Y(t) = -t - \frac{1}{2}.$$

Hence, the general solution of (1) is

$$\begin{aligned} y(t) &= y_c(t) + Y(t) \\ \text{or, } y(t) &= c_1 e^t + c_2 e^{-2t} - t - \frac{1}{2} \quad \dots \quad (3) \end{aligned}$$

$$\text{or, } y'(t) = c_1 e^t - 2c_2 e^{-2t} - 1 \quad \dots \quad (4)$$

Using $y(0) = 0$ in (3), we get

$$0 = c_1 + c_2 - 0 - \frac{1}{2}$$

$$\text{or, } c_1 + c_2 = \frac{1}{2}$$

$$\text{or, } 2c_1 + 2c_2 = 1 \quad \dots \quad (5)$$

Again, using $y'(0) = 1$ in (4), we get

$$1 = c_1 - 2c_2 - 1$$

$$\text{or, } c_1 - 2c_2 = 2 \quad \dots \quad (6)$$

Solving (5) and (6), we get

$$c_1 = 1, \quad c_2 = -\frac{1}{2}$$

Hence, the required solution of the given IVP is

$$y(t) = e^t - \frac{1}{2}e^{-2t} - t - \frac{1}{2}.$$

Example 2. Find the general solution of the given initial value problem $y'' + 4y = t^2 + 3e^t, y(0) = 0, y'(0) = 2$.

Solution

The given differential equation is

$$y'' + 4y = t^2 + 3e^t \quad \dots \quad (1)$$

with initial conditions

$$y(0) = 0, y'(0) = 2.$$

The A.E. is

$$m^2 + 4 = 0$$

$$\therefore m = \pm 2i$$

Thus, the complementary solution is

$$y_c(t) = c_1 \cos 2t + c_2 \sin 2t \quad \dots \quad (2)$$

A particular solution of (1) may be chosen as

$$Y(t) = (At^2 + Bt + c) + De^t$$

Then,

$$Y'(t) = 2At + B + De^t$$

and

$$Y''(t) = 2A + De^t$$

Using these values in (1), we get

$$2A + De^t + 4[(At^2 + Bt + c) + De^t] = t^2 + 3e^t$$

$$\text{or, } 4At^2 + 4Bt + 5De^t + (2A + 4C) = t^2 + 3e^t$$

Equating the coefficients of like terms, we get

$$4A = 1, 4B = 0, 5D = 3, 2A + 4C = 0$$

$$\Rightarrow A = \frac{1}{4}, B = 0, D = \frac{3}{5}, C = -\frac{1}{8}$$

Thus,

$$Y(t) = \frac{1}{4}t^2 + 0.t - \frac{1}{8} + \frac{3}{5}e^t$$

$$\text{or, } Y(t) = \frac{1}{4}t^2 - \frac{1}{8} + \frac{3}{5}e^t.$$

Hence, the general solution of (1) is

$$y(t) = y_c(t) + Y(t)$$

$$\Rightarrow y(t) = c_1 \cos 2t + c_2 \sin 2t + \frac{1}{4}t^2 - \frac{1}{8} + \frac{3}{5}e^t \quad \dots \quad (3)$$

$$\Rightarrow y'(t) = -2c_1 \sin 2t + 2c_2 \cos 2t + \frac{1}{2}t + \frac{3}{5}e^t \quad \dots \quad (4)$$

Using $y(0) = 0$, (3) gives

$$\begin{aligned} 0 &= c_1 \cdot 1 + c_2 \cdot 0 + \frac{1}{4} \cdot 0 - \frac{1}{8} + \frac{3}{5} \cdot 1 \\ \Rightarrow c_1 &= -\frac{19}{40} \end{aligned}$$

Again, using $y'(0) = 2$, (4) gives

$$\begin{aligned} 2 &= -2c_1 \cdot 0 + 2c_2 \cdot 1 + \frac{1}{2} \cdot 0 + \frac{3}{5} \cdot 1 \\ \Rightarrow c_2 &= \frac{7}{10} \end{aligned}$$

Hence, the required particular solution of (1) is

$$y(t) = -\frac{19}{40} \cos 2t + \frac{7}{10} \sin 2t + \frac{1}{4}t^2 - \frac{1}{8} + \frac{3}{5}e^t.$$

Example 3. Find the general solution of the given initial value problem $y'' + 4y = 2\sin 2t$, $y(0) = 2$, $y'(0) = -1$

Solution

The given equation is

$$y'' + 4y = 2\sin 2t \quad \dots \quad (1)$$

with initial conditions

$$y(0) = 2, y'(0) = -1$$

The A.E. is

$$\begin{aligned} m^2 + 4 &= 0 \\ \therefore m &= -2i \end{aligned}$$

Therefore, the complementary solution is

$$y_c(t) = c_1 \cos 2t + c_2 \sin 2t \quad \dots \quad (t)$$

For particular solution, let

$$Y(t) = t(A \cos 2t + B \sin 2t) = At \cos 2t + Bt \sin 2t$$

Then,

$$\begin{aligned} Y'(t) &= A(\cos 2t - 2t \sin 2t) + B(\sin 2t + 2t \cos 2t) \\ &= A \cos 2t - 2At \sin 2t + B \sin 2t + 2Bt \cos 2t \\ Y''(t) &= -2A \sin 2t - 2A(\sin 2t + 2t \cos 2t) + 2B \cos 2t + 2B(\cos 2t - 2t \sin 2t) \\ &= -4A \sin 2t - 4At \cos 2t + 4B \cos 2t - 4Bt \sin 2t \end{aligned}$$

Substituting these values in (1), we get

$$\begin{aligned} -4A \sin 2t - 4At \cos 2t + 4B \cos 2t - 4Bt \sin 2t + 4At \cos 2t + 4Bt \sin 2t &= 2 \sin 2t \\ -4A \sin 2t + 4B \cos 2t &= 2 \sin 2t \end{aligned}$$

Equating the coefficients of like terms, we get

$$\begin{aligned} -4A &= 2, 4B = 0 \\ \Rightarrow A &= -\frac{1}{2}, B = 0 \end{aligned}$$

Therefore, the particular solution is

$$Y(t) = t \left(-\frac{1}{2} \cos 2t + 0 \times \sin 2t \right) = -\frac{1}{2} t \cos 2t.$$

Thus, the general solution is

$$y(t) = y_c(t) + Y(t)$$

$$y(t) = c_1 \cos 2t + c_2 \sin 2t - \frac{1}{2} t \cos 2t \quad \dots \quad (3)$$

Also,

$$y'(t) = -2c_1 \sin 2t + 2c_2 \cos 2t - \frac{1}{2} \cos 2t + t \sin 2t \quad \dots \quad (4)$$

Using $y(0) = 2$, then (3) gives

$$2 = c_1 + 0 - 0 \therefore c_1 = 2$$

Using $y'(0) = -1$, equation (4) gives

$$-1 = 0 + 2c_2 - \frac{1}{2} = 0 \Rightarrow c_2 = -\frac{1}{4}.$$

Hence, the required solution of the given IVP is

$$y(t) = y_c(t) + Y(t) = 2 \cos 2t - \frac{1}{4} \sin 2t - \frac{1}{2} t \cos 2t.$$

Variation of Parameters

Theorem (Only Statement)

Suppose that the functions p, q and g are continuous on an open interval I and the functions y_1 and y_2 form a fundamental set of solutions of the homogeneous equation

$$y'' + p(t)y' + q(t)y = 0 \quad \dots \quad (1)$$

corresponding to the non-homogeneous equation

$$y'' + p(t)y' + q(t)y = g(t) \quad \dots \quad (2)$$

Then, a particular solution of (2) is

$$Y(t) = -y_1(t) \int_{t_0}^t \frac{y_2(s)g(s)}{W(y_1, y_2)(s)} ds + y_2(t) \int_{t_0}^t \frac{y_1(s)g(s)}{W(y_1, y_2)(s)} ds$$

where t_0 is any conveniently chosen point in the interval I . The general solution is

$$y = c_1 y_1(t) + c_2 y_2(t) + Y(t).$$

Example 1.

Find the general solution of $y'' - y' - 2y = 2e^{-t}$ by using the method of variation of parameters.

Solution

The given de is

$$y'' - y' - 2y = 2e^{-t} \quad \dots \quad (1)$$

It's A.E. is

$$\begin{aligned} m^2 - m - 2 &= 0 \\ \Rightarrow m &= -1, 2. \end{aligned}$$

Therefore,

$$y_c(t) = c_1 e^{-t} + c_2 e^{2t}$$

is a complementary solution of (1).

Here,

$$y_1(t) = e^{-t} \text{ and } y_2(t) = e^{2t}$$

form a fundamental set of solutions of the corresponding homogeneous equation.

Now,

$$y_1'(t) = -e^{-t} \text{ and } y_2'(t) = 2e^{2t}$$

Then,

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{vmatrix} = 3e^t \neq 0.$$

We have $g(t) = 2e^{-t}$.

Then,

$$\begin{aligned}
Y(t) &= -y_1(t) \int \frac{y_2(t)g(t)}{W(y_1, y_2)(t)} dt + y_2(t) \int \frac{y_1(t)g(t)}{W(y_1, y_2)(t)} dt \\
&= -e^{-t} \int \frac{e^{2t} \times 2e^{-t}}{3e^t} dt + e^{2t} \int \frac{e^{-t} \times 2e^{-t}}{3e^t} dt \\
&= -\frac{2}{3} e^{-t} \int dt + \frac{2}{3} e^{2t} \int e^{-3t} dt \\
&= -\frac{2}{3} e^{-t} \times t + \frac{2}{3} e^{2t} \times \left(\frac{e^{-3t}}{-3} \right) \\
&= -\frac{2}{3} t e^{-t} - \frac{2}{9} e^{-t}
\end{aligned}$$

Therefore,

$$Y(t) = -\frac{2}{3} t e^{-t} - \frac{2}{9} e^{-t}.$$

is a particular solution of (1). Hence, the required general solution of (1) is

$$y(t) = y_c(t) + Y(t) = c_1 e^{-t} + c_2 e^{2t} - \frac{2}{3} t e^{-t} - \frac{2}{9} e^{-t}$$

Example 2.

Verify that $y_1(t) = t^2, y_2(t) = t^{-1}$ satisfy the corresponding homogeneous differential equation and then find a particular solution of non-homogeneous ODE $t^2 y'' - 2y = 3t^2 - 1, t > 0$.

Solution

The given ODE is

$$t^2 y'' - 2y = 3t^2 - 1, t > 0 \quad \dots \quad (1)$$

$$\text{or,} \quad y'' - \frac{2}{t^2} y = \frac{3t^2 - 1}{t^2}, t > 0$$

Here, $g(t) = \frac{3t^2 - 1}{t^2}$.

The corresponding homogeneous equation of (1) is

$$t^2 y'' - 2y = 0 \quad \dots \quad (2)$$

We take $y_1(t) = t^2$ as one of the solutions.

Therefore,

$$y_1'(t) = 2t, \quad y_1''(t) = 2.$$

Putting in (2), we get

$$t^2 \times 2 - 2t^2 = 0 \Rightarrow 0 = 0$$

Therefore, $y_1(t) = t^2$ satisfies homogeneous equation (2)

Again, consider the next solution $y_2(t) = t^{-1}$. Then,

$$y_2'(t) = -t^{-2} \text{ and } y_2''(t) = 2t^{-3}.$$

Substituting in (2), we get

$$t^2 \times 2t^{-3} - 2t^{-1} = 0 \Rightarrow 2t^{-1} - 2t^{-1} \Rightarrow 0 = 0$$

Therefore, $y_2(t) = t^{-1}$ also satisfies the homogeneous equation (2).

Also,

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} t^2 & t^{-1} \\ 2t & -t^{-2} \end{vmatrix} = -1 - 2 = -3 \neq 0$$

Thus, y_1 and y_2 form a fundamental set of solutions of (2).

Next, the particular solution is

$$Y(t) = -y_1(t) \int \frac{y_2(t)g(t)}{W(y_1, y_2)(t)} dt + y_2(t) \int \frac{y_1(t)g(t)}{W(y_1, y_2)(t)} dt$$

$$\begin{aligned}
&= -t^2 \int \frac{t^{-1} \times \frac{3t^2 - 1}{t^2}}{-3} dt + t^{-1} \int \frac{t^2 \times \frac{3t^2 - 1}{t^2}}{-3} dt \\
&= \frac{t^2}{3} \int \left(\frac{3}{t} - t^{-3} \right) dt - \frac{t^{-1}}{3} \int (3t^2 - 1) dt \\
&= \frac{t^2}{3} \left[3 \ln t + \frac{t^{-2}}{2} \right] - \frac{t^{-1}}{3} [t^3 - t] \\
&= t^2 \ln t + \frac{1}{6} - \frac{t^2}{3} + \frac{1}{3}.
\end{aligned}$$